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Further Paleomagnetic Results From the Builth Wells Ordovician Inlier, Wales

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A characteristic magnetization has been isolated from 18 sites in Middle Ordovician igneous and sedimentary rocks from the Builth inlier, Wales. Two conglomerate tests demonstrate that the characteristic magnetization in two of the volcanic units was acquired early, perhaps during initial emplacement. Six sites from the northern and central parts of the inlier give a mean direction of D=153.8°, I=69.8° (k=45.8; α_{95}=10°) that passes a regional fold test. However, 12 sites from the southern end of the inlier, in and near the Llanelwedd quarry, yield a significantly different direction after correction for local tilt: D=176.4°, I=44.5° (k=42.4; α_{95}=6.7°). The Llanelwedd quarry data are anomalous with respect to both results from the northern parts of the inlier, and those obtained from approximately coeval rocks from the Shelve inlier. We have searched for an explanation for the discrepancy between the magnetization directions of the two areas and have now ruled out some possibilities, including superimposed magnetization components at one of the two areas. The Llanelwedd quarry lies within a wedge of dominantly andesitic volcanic rocks that appear only locally, and which are overlain by two unconformities that are also not present elsewhere in the inlier. One possible explanation for the anomalous nature of the Llanelwedd quarry magnetization is that the strata were not horizontal at the time of magnetization, either due to early volcano-tectonic tilting or primary dips on the flanks of a volcano. Another possibility is that the Llanelwedd quarry rocks were extruded rapidly enough that the geocentric coaxial dipole field hypothesis cannot be applied. Results from Cenozoic volcanic rocks from the western United States suggest that even highly anomalous fields can be preserved virtually unchanged over a number of separate flow events. We currently regard this to be the most viable explanation for the anomalous nature of the Llanelwedd quarry magnetization. We conclude that the data from the 6 northern sites represent the best estimate of the Middle Ordovician axial dipole field for the inlier.

Introduction

There is currently much interest in the closure history of the Iapetus Ocean in the British sector. Analysis of shelf faunas suggests a wide Early Ordovician separation between Laurentia (including northern Scotland) on the one hand, and southern Britain and Gondwana on the other [*Cocks and Fortey*, 1982]. By Middle Ordovician, there was faunal divergence between Gondwana and southern Britain [*Cocks and Fortey*, 1982], and progressive convergence of faunas in southern Britain and Laurentia until the Early Devonian [*McKerrow and Cocks*, 1976]. Provenance links [*Murphy and Hutton*, 1986], sediment source arguments [*Soper and Woodcock*, 1990], and tectonic considerations [*Kneller*, 1991] suggest initial continental contact between southern Britain and Laurentia by Wenlock time. Final suturing is generally regarded as being coeval with Acadian (Late Caledonian) deformation in the Early Devonian.

The history of Ordovician and Silurian convergence between southern Britain and Laurentia has not been well constrained by high-quality paleomagnetic data until very recently [see *Channell et al.*, 1992a for review]. For the Middle Ordovician of southern Britain, the published studies incorporating both stepwise demagnetization and field tests to constrain the magnetization age yield a small, but in this case highly significant, discrepancy in inclination, and therefore also in paleolatitude. *Briden and Mullan* [1984] and *Trench et al.* [1991] have obtained very similar results yielding a paleolatitude of about 35° south, both constrained by conglomerate tests, from Upper Llanvirn lavas and associated dolerites from the southern portion of the Builth inlier in Wales. However, *McCabe and Channell* [1990] have presented a result from Lower Llanvirn rocks from the Shelve inlier which is constrained to the Ordovician by a regional fold test, but which gives a paleolatitude of about 51° south. The 16° paleolatitude difference between the nearly coeval Builth and Shelve paleomagnetic data results in a 1600 km difference in the estimate of the Middle Ordovician latitudinal width of the Iapetus Ocean. It is therefore important to try to find the origin of the discrepancy between the Builth and Shelve results.

With this objective in mind, we embarked on the present study to extend sampling into the central and northern reaches of the Builth inlier. Preliminary results [*McCabe and Channell*, 1991] confirmed the positive conglomerate tests, and the previously reported moderate inclinations (and paleolatitudes) for the southern part of the inlier. However, we found steeper, "Shelve-like" inclinations (higher paleolatitudes) in two sites in the northern part of the inlier [*McCabe and Channell*, 1991]. We present here a more fully-developed data base from the Builth inlier that bears upon the Builth/Shelve paleolatitude discrepancy.

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GEOLOGICAL SETTING AND SAMPLING

The Builth Wells-Llandrindod Wells Ordovician inlier (herein called the Builth inlier) is one of the classic areas of British geology (Figure 1). Early studies by *Murchison* [1833, 1839] and *de la Beche* [1846] established the stratigraphic age and general depositional setting of the Ordovician rocks within the inlier. Paleontological and stratigraphic studies were reviewed by *Elles* [1939], and the area was mapped in great detail (at a scale of 25 inches to the mile) by *Jones and Pugh* [1941, 1946, 1948, 1949]. The *Institute of Geological Sciences* [1977] has published a 1:25,000 geological map of the area. The inlier exposes Llanvirn marine sedimentary rocks and associated lavas and tuffs which are overlain by marine shales of Llandeilo age (Figure 2). Dolerites of Ordovician age are found as mainly concordant intrusions throughout the inlier [*Jones and Pugh*, 1948]. The bedded rocks in the western portion of the inlier dip 25 to 40° to the northwest. Further east, open NE trending folds with sub-horizontal axes are present. A system of steep NE and E trending faults with apparent dextral offset affect the Ordovician rocks of the inlier [*Woodcock*, 1987]. The Builth inlier lies to the southeast of the Caledonian front and to the north of the Variscan front (Figure 1); thus folding and cleavage of Acadian and Hercynian age are very weak. The deformation is mainly of latest Ordovician (Ashgill) age [*Woodcock and Gibbons*, 1988], and is manifest as tilting and minor folding associated with dextral strike-slip faulting [*Woodcock*, 1987]. Some minor Silurian or post-Silurian reactivation of older structures can be observed in the Silurian cover of the inlier [*Woodcock*, 1987].

The southern area of the inlier contains a wedge of northwest-dipping upper Llanvirn lavas that reaches a maximum thickness of some 200 m in the vicinity of the Llanelwedd quarry near Builth Wells, but which thins northward to 0 thickness over a distance of about 3 km (Figures 2 and 3). The Llanelwedd quarry exposes spilitic lava flows which contain horizons of thin "felsite agglomerate" [*Jones and Pugh*, 1949], which have been interpreted as ignimbrite and lahar deposits [*Furnes*, 1978]. Keratophyre flows overlie the spilites (Figure 3). This volcanic package is succeeded unconformably by a shoreline deposit, the Newmead Sandstone and associated boulder beds, which in turn is overlain by Llandeilo marine shales with angular discordance [*Elles*, 1939; *Jones and Pugh*, 1949]. This distinctive volcanic assemblage and the two unconformities that immediately succeed it are not present in correlative rocks further north in the inlier, where the Middle Ordovician section appears to be a conformable marine sequence (see Figure 2, columns *a* and *b*). The large boulders found near the base of the Newmead Sandstone are largely derived from the underlying keratophyres. They are associated with an erosional surface including steep cliff faces cut into the underlying rocks. These features and the presence of marine fossils indicate that the Newmead was deposited adjacent to wave-cut cliffs in a beach-shoreface setting [*Jones and Pugh*, 1949]. A similar boulder bed can be found within a "spilite boulder tuff" near the base of the volcanic pile in the eastern part of the Llanelwedd quarry.

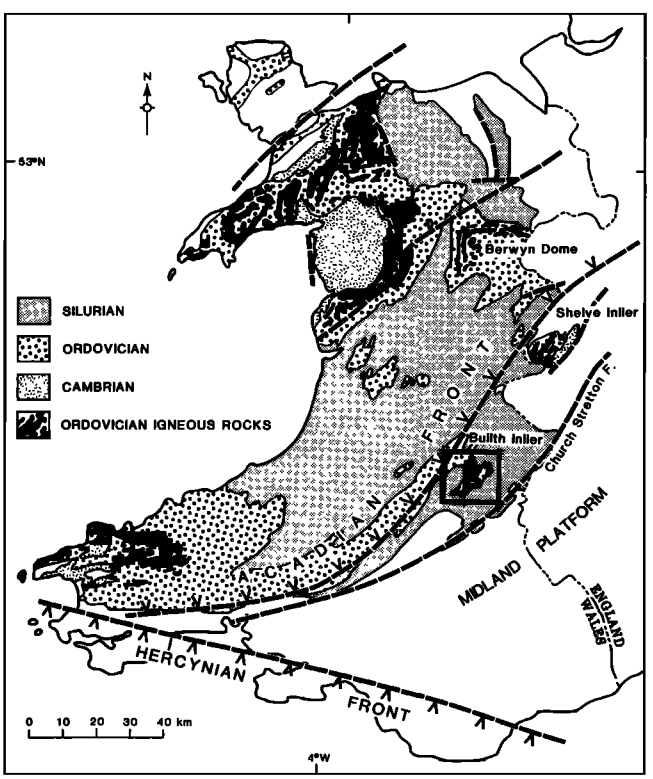


Fig. 1. Generalized geological map of Wales [after *Kokelaar et al.*, 1984] showing the location of the Builth inlier.

We sampled lavas, intrusives, and sedimentary rocks at 31 sites distributed throughout the Builth inlier (Table 1) using a gasoline-powered portable rock drill. Intrusive bodies were sampled only if bedded country rock was exposed, and the intrusions appeared to be sufficiently concordant for the estimation of paleohorizontal using the attitude of the intruded sedimentary rocks. Core samples were oriented with a magnetic compass and clinometer. Care was taken to sample strata with differing bedding attitudes for a regional paleomagnetic fold test. In addition, we sampled clasts within a "felsite agglomerate", the "beach facies" boulders within the Newmead Sandstone, and in the "spilite boulder tuff" found beneath the Llanelwedd volcanic pile for purposes of the paleomagnetic conglomerate test.

Our preliminary paleomagnetic results from the Builth inlier indicated a discrepancy between paleomagnetic directions from the Llanelwedd quarry and the rocks exposed further north in the inlier [*McCabe and Channell*, 1991]. Further investigation has confirmed that discrepancy. Therefore we will treat the results from the Llanelwedd quarry and its immediate vicinity (the "southern" sites) separately from results obtained north of the outcrop area of the Llanelwedd volcanics (the "northern" sites). The locations of sampled sites are given in Table 1. Sites which yielded resolvable characteristic magnetization components are plotted in Figures 2 and 3.

LABORATORY METHODS

The samples we collected from the Builth inlier were divided into two subequal groups. About half of the samples went to the University of Florida for laboratory treatment, and the other half to Louisiana State University (Table 1). Core samples were cut into standard 2.2 cm specimens for paleomagnetic analysis. Pilot demagnetization studies indicated that thermal demagnetization resulted in better separation of magnetization components for most samples, but alternating field (AF) demagnetization was found to be satisfactory for samples of marine siltstone from site LI 2. All samples were subjected to stepwise demagnetizations in at least 13 steps, but most of the samples were treated in 20 or more steps. At the University of Florida,

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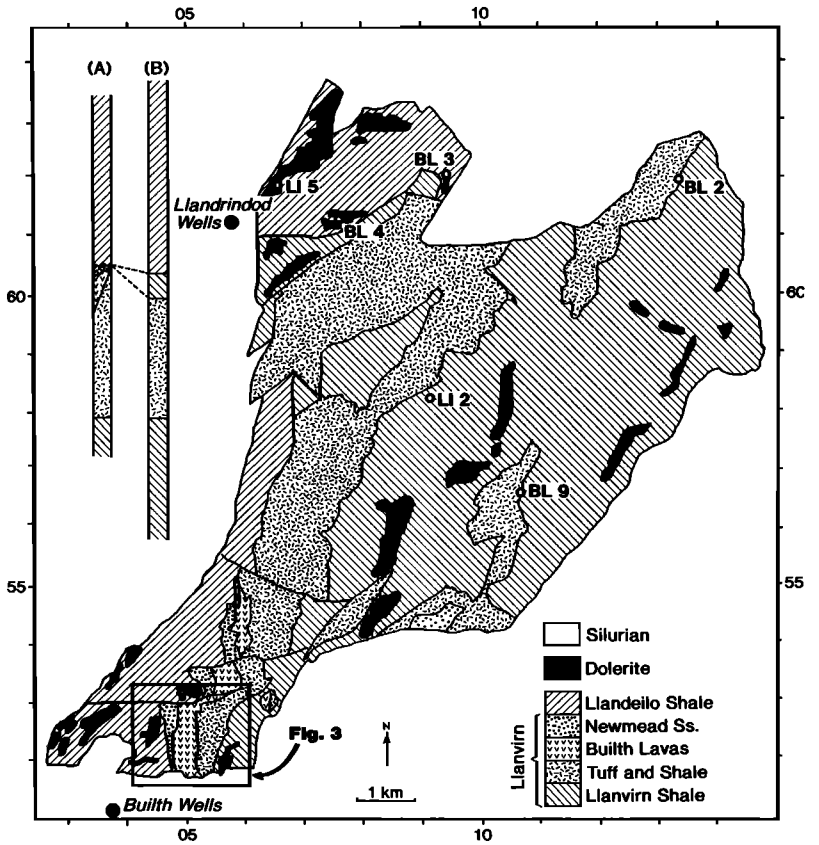


Fig. 2. Generalized geological map of the Builth Ordovician inlier showing the locations of sampling sites in the northern and central reaches of the inlier. The sites shown are those that yielded a characteristic magnetization after demagnetization. The southern area of the inlier is within the square, and is shown in more detail in Figure 3. A generalized stratigraphic column for (*a*) the area near Builth Wells is compared to [*b*) one near Llandrindod Wells. Note that the unconformities succeeding the lavas near Builth Wells are not present in correlative rocks from the area to the north [after *Institute of Geological Sciences*, 1977].

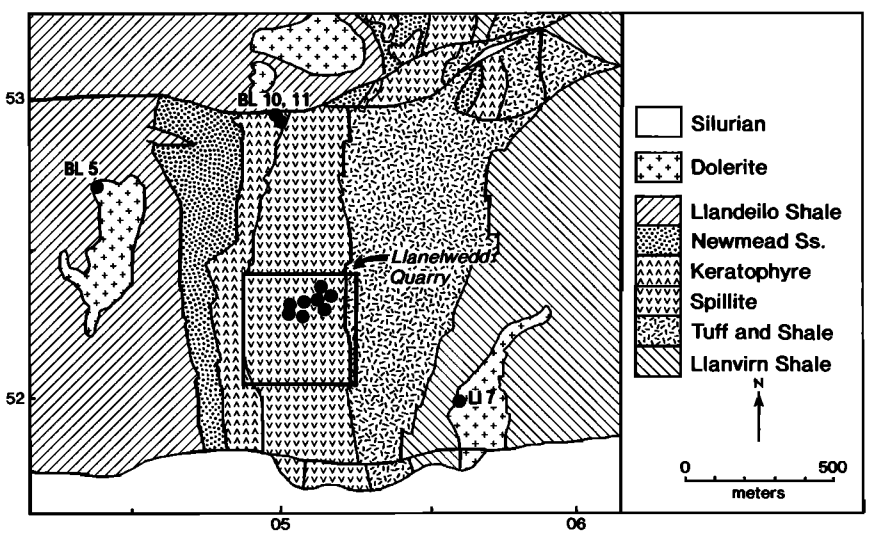


Fig. 3. Generalized geological map of the southern area of the Builth inlier. Sites that yielded a characteristic magnetization are shown. The sites within the Llanelwedd quarry are unlabeled for clarity. They are sites LI 9, LI 11, LI 12, LI 13, LI 15, LI 16, LI 18, and LI 19 [after *Woodcock*, 1987; and *Institute of Geological Sciences*, 1977].

samples were heated in a Schonstedt furnace, and remanent magnetizations measured with a two-axis ScT cryogenic magnetometer. At Louisiana State University, samples were thermally treated in a three zone furnace constructed from Lindberg components, or AF treated with a tuned solenoid driven by a Behlman AC power supply. Remanent magnetization were measured with a three-axis superconducting magnetometer manufactured by CTF Systems.

Demagnetization data were plotted on orthogonal demagnetization diagrams [*Zijderveld*, 1967], and component magneti-

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TABLE 1. Site Data, Including Lithology, Location, Bedding Attitude, Where the Samples Were Demagnetized, and the Number of Component Directions Used in the Data Analysis for the Secondary Magnetization (Ns) and Characteristic Magnetization (Np)

Site #	Lithology	UF or LSU	Grid Ref (SO)	S/D	Ns/No	N p/No
LI 1	Marine seds (*D. bifidus* zone)	L	097 550	15/36	3/15*	0/15
LI 2	Marine seds (*D. bifidus* zone)	L	091 582	224/16	0/11	11/11
LI 3	Tuffs (*D. murchisoni* zone)	F	093 590	198/33	0/12	0/12
LI 4	Tuffs (*D. murchisoni* zone)	L	092 591	208/42	0/14	0/14
LI 5	Dolerite	L	065 618	237/34	0/16	11/16
LI 6	Tuffs (*D. murchisoni* zone)	L	091 596	11/60	8/11	0/11
LI 7	Dolerite	F	056 520	181/50	0/15	15/15
LI 9	Spilite	F	050 524	162/34	14/14	14/14
LI 10	Spilite	L	050 523	159/40	14/14	6/14*
LI 11	Spilite	F	052 523	179/37	12/15	15/15
LI 12	Spilite	L	051 523	155/37	3/14*	14/14
LI 13	Spilite	F	050 523	164/33	14/15	15/15
LI 14	Spilite	L	052 522	164/33	0/14	0/14
LI 15	Spilite	F	051 524	153/34	2/14*	14/14
LI 16	Spilite	L	052 524	162/31	12/13	8/13
LI 18	Spilite	F	051 523	162/31	10/11	11/11
LI 19	Spilite	F	051 523	160/34	12/16	16/16
BL 1	Dolerite	F	056 669	209/56	0/14	0/14
BL 2	Keratophyre	L	133 619	214/38	0/14	13/14
BL 3	Dolerite	F	094 620	324/38	0/28	28/28
BL 4	Dolerite	L	076 612	257/45	0/15	14/15
BL 5	Dolerite	F	044 527	162/42	0/16	16/16
BL 6	Dolerite	L	030 523	211/47	0/13	0/13
BL 7	Tuffs (*D. murchisoni* zone)	F	094 549	144/14	13/19	0/19
BL 8	Marine seds (*D. bifidus* zone)	F	097 557	14/33	0/16	0/16
BL 9	Tuffs (*D. murchisoni* zone)	F	107 566	201/48	14/15	8/15
BL 10	Keratophyre	F	049 529	159/26	0/14	14/14
BL 11	Keratophyre	F&L	049 529	189/30	0/19	19/19
BL 12	Sandstone	L	047 528	194/32	0/22	0/22
BL 13	Sandstone	F	047 527	189/25	14/16	0/16
BL 17	Dolerite	F	030 525	174/40	0/16	0/16

S is bedding strike and D is bedding dip (dip direction is clockwise from strike); L, Louisiana State University; F, University of Florida; No is number of specimens demagnetized; 71 specimens (31 sites) were subjected to stepwise demagnetization; 262 characteristic specimen directions were obtained from 19 sites (18 pass 50% criterion); 145 secondary directions from 14 sites (11 pass 50% criterion).
*Not used (fails 50% criterion).

zations were calculated using principal component analysis [*Kirschvink*, 1980] after careful visual inspection of the diagrams. A magnetization was judged to have been sufficiently resolved if three or more data points followed a linear trend. As is usually the case for Paleozoic rocks, many of the samples yielded data that were too noisy for reliable determination of magnetization directions using this method. Site data for a magnetization component were accepted for the final data analysis if 50% of the samples demagnetized yielded resolvable data for that component. By this "50% criterion", a characteristic site mean magnetization was obtained from 18 of 31 sampled sites; an ancient secondary magnetization was resolved in 11 sites. Site mean and formation mean paleomagnetic directions were calculated using *Fisher* [1953] statistics.

RESULTS

Northern Sites

Only 6 of 13 of the northern sites yielded a resolvable characteristic magnetization. Better success was hampered by the general lack of fresh, unweathered lavas in this part of the inlier. Many of the sites sampled were in marine tuffs and marine sedimentary rocks, and only two of eight sites sampled in these lithologies gave interpretable data. The remaining sites were in dolerite or keratophyre; all but one site in these lithologies yielded demagnetization data of high quality. An overprint of probable Late Paleozoic age was resolved in only one of the northern sites. Typical demagnetization diagrams for the six good northern sites are shown in Figure 4. A southerly to southeasterly and steeply down characteristic magnetization (after tilt correction) was found in most samples from these sites. The unblocking temperature range varies in the different lithologies, but maximum unblocking temperatures above 585°C were not observed, suggesting magnetite as the magnetization carrier. Some samples have essentially single component magnetizations (e.g. Figures 4*a* and 4*b*). Others have an overprint of probable viscous origin (Fig. 4*c*). An overprint of probable Late Paleozoic age was found in site BL 9 (Figure 4*d*).

Specimen characteristic magnetizations are shown in Figure 5 (right) in coordinates corrected for local tilt. Site mean characteristic magnetization directions and pertinent statistical parameters are given in Table 2 and plotted in Figure 6. As can be seen in these data, the site mean magnetization directions cluster much better after correction for local tilt, indicating that the magnetization was acquired prior to late Ordovician folding. The paleomagnetic fold test is positive at the 99% confidence level [*McElhinny*, 1964; *McFadden*, 1990]. The mean direction after tilt correction is D = 153.8, I = 69.8 (k = 45.8, α_{95} = 10°), yielding a paleolatitude for the site of 54°. This result is very similar to the mean direction we previously obtained for

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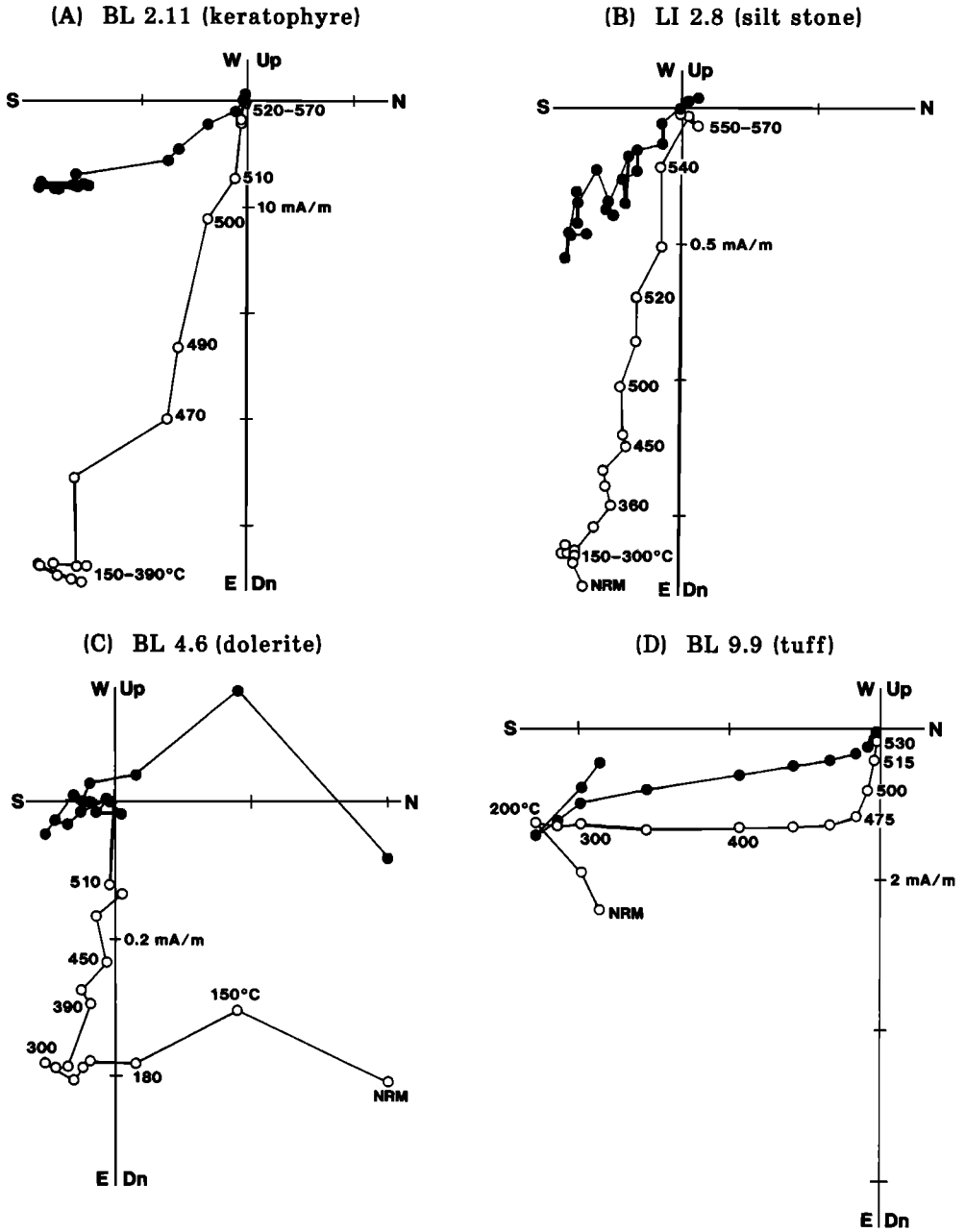


Fig. 4. Thermal demagnetization data for typical specimens from the northern sites of the Builth inlier. Data are corrected for local tilt. Open circles are projections of the magnetization vectors onto the vertical plane, solid circles are projections onto the horizontal plane.

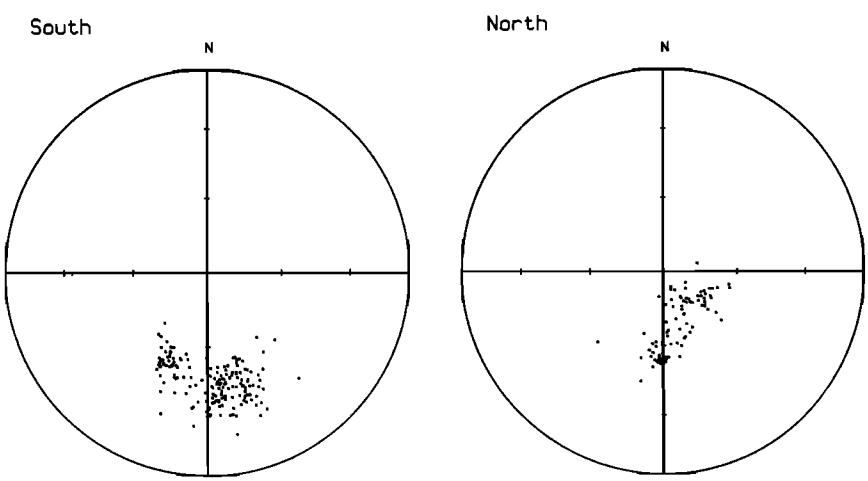


Fig. 5. Specimen characteristic magnetization directions in equal area projection for the southern, and northern sites in the Builth inlier, in coordinates corrected for local tilt. All directions plot on the lower hemisphere.

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TABLE 2. Site Mean Characteristic Magnetization Data from the Central and Northern Sites of the Builth-Llandrindod Inlet

Site	N	k	α_{95}	In Situ Dec	In Situ Inc	Tilt Corrected Dec	Tilt Corrected Inc
LI 2	11	288.6	2.7	126.0	54.3	120.2	70.0
LI 5	11	104.6	4.5	157.1	33.2	168.3	66.2
BL 2	13	97.5	4.2	128.9	34.4	137.2	72.1
BL 3	28	240.4	1.8	204.1	25.1	182.4	54.8
BL 4	14	40.0	6.4	166.7	31.5	165.9	76.5
BL 9	8	229.6	3.7	118.0	21.8	129.4	69.0
Means	6	7.8	25.6	150.7	37.2		
	6	45.8	10.0			153.8	69.8
Pole (bedding corrected):						17.9° N;	2.9° E
						$\delta p = 14.7$	
						$\delta m = 17.2$	

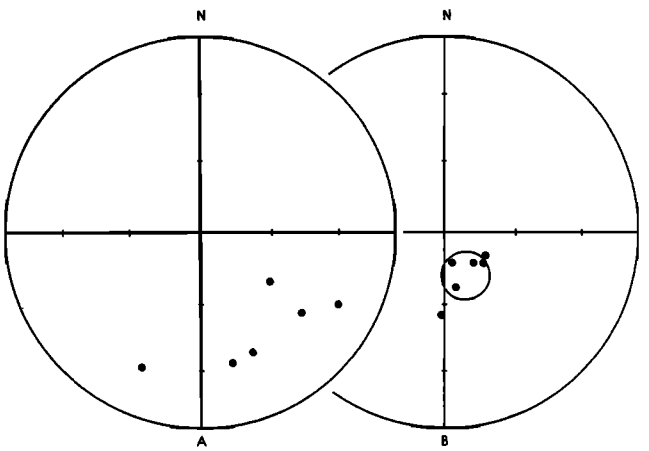


Fig. 6. Site mean characteristic directions for the central and northern sites in equal area projection, (*a*) before bedding correction, and (*b*) after correction for local tilt. The 95% circle of confidence about the mean is shown in the tilt corrected plot (the mean direction itself is omitted for clarity).

Middle Ordovician volcanic rocks from the nearby Shelve inlier [*McCabe and Channell*, 1990], which yielded a tilt corrected mean direction of D = 116.4, I = 67.9. In both studies, regional fold tests constrain the magnetization age to the Ordovician, all magnetizations are of reversed polarity, and the paleolatitude is in the range of 51 to 54° south.

Southern Sites

We previously reported the results of conglomerate tests on clasts from a "felsite agglomerate" exposed within the Llanelwedd quarry, and combined results from the "beach facies" boulders from the Newmead Sandstone, and from the "spilite boulder tuff" beneath the Llanelwedd volcanic pile [*McCabe and Channell*, 1991]. In all cases, different samples from the same clast gave the same characteristic magnetization direction, but the clast mean directions were randomly distributed, indicating that the clasts were magnetized before deposition of the conglomeratic beds. Those data are shown again in Figure 7. Our results from the "felsite agglomerate" (Figure 7*a*) are fully compatible

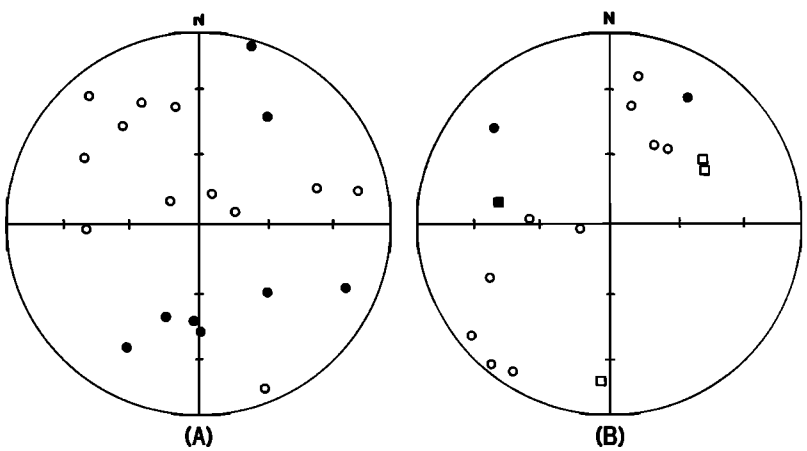


Fig. 7. (*a*) Clast mean directions for the "felsite agglomerate" conglomerate test. Directions from the same clast were well clustered. Based on data from 30 samples taken from 20 clasts. Conventions as in Figure 4. (*b*) Combined clast mean directions for boulders from the "beach facies" conglomerates (circles), the "spilite boulder tuff" (squares). Directions from the same clast were extremely well clustered. Based on data from 43 samples taken from 16 clasts. Open symbols are on the upper hemisphere, solid symbols the lower hemisphere. Equal area projections.

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with those of *Briden and Mullan* [1984], which were based on a much smaller collection. The positive conglomerate test from the large boulders (Figure 7*b*) is similar to the result obtained from Newmead Sandstone boulders by *Trench et al.* [1991]. There can be little doubt that the Llanelwedd quarry spilites and keratophyres contain a magnetization that dates to initial emplacement of these volcanic units.

Typical demagnetization diagrams from in-place spilite flows sampled in the Llanelwedd quarry are shown in Figures 8*a* and 8*b*. A southerly and shallow magnetization overprint of probable Kiaman superchron age is present in nearly all quarry samples. A higher unblocking temperature characteristic magnetization with southerly declinations and moderate downward inclinations after local tilt correction is also present in most of the samples. We had previously noted [*McCabe and Channell*, 1991] that there appeared to be some overlap in the unblocking spectra of the two magnetization components, which might explain the fact that the inclinations observed in the quarry are shallower that those found in the northern Builth sites and in the Shelve inlier [*McCabe and Channell*, 1990]. However, *Trench et al.* [1991] noted that the Llanelwedd keratophyres show a similarly shallow characteristic magnetization, but no evidence for an overprint of Late Paleozoic age. Our own data from two keratophyre sites confirms this (Figure 8*d*). It appears therefore that an alternative explanation must be sought for the difference in inclination between the northern and southern Builth sites. Two dolerite sites from the southern end of the inlier also give a similar characteristic direction after correction for tilt of the surrounding shale beds (Figure 8*c*). Specimen characteristic magnetization directions from the southern sites

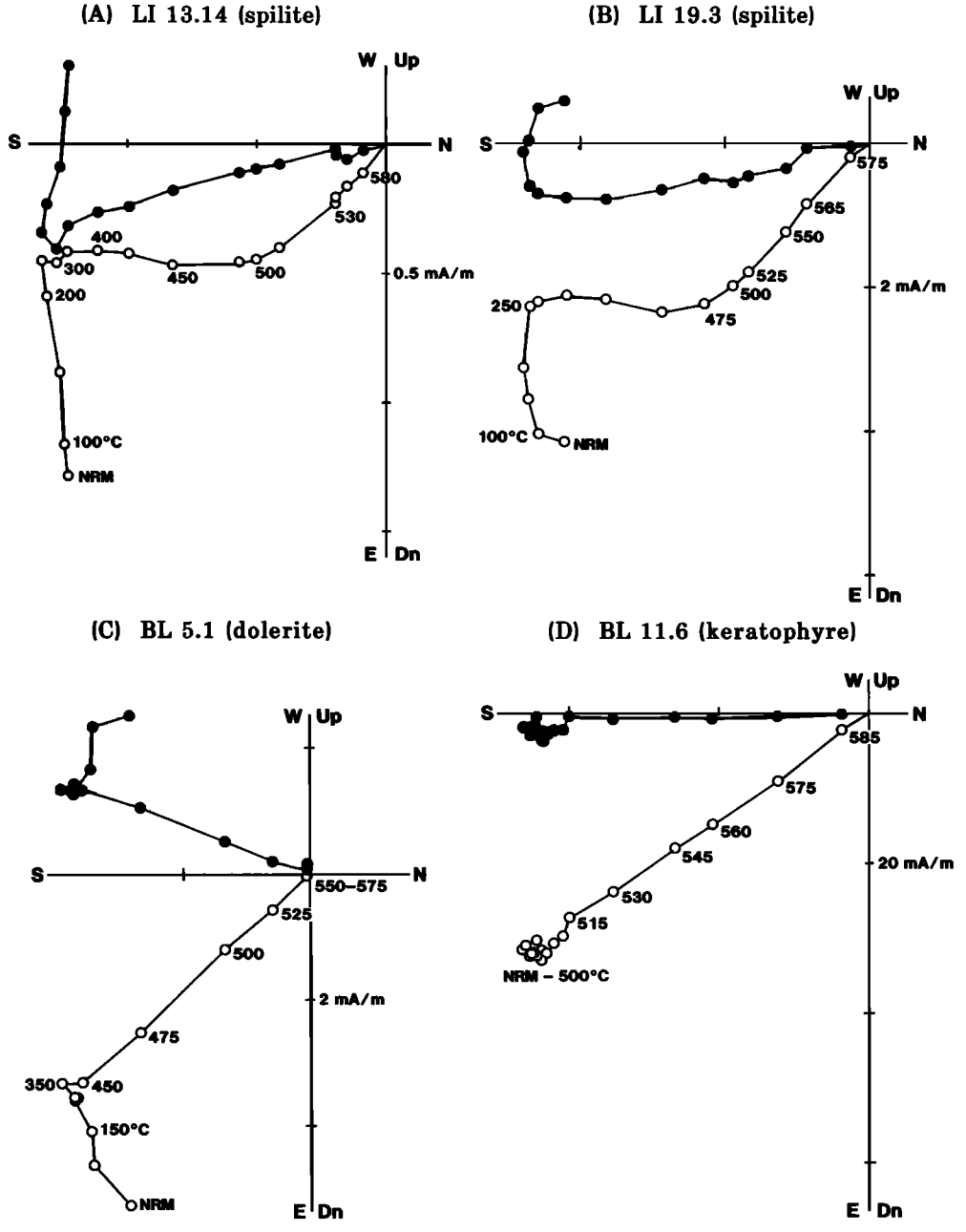


Fig. 8. Thermal demagnetization data for typical samples of in-place rock from the Llanelwedd quarry, and other nearby "southern" sites. Data are corrected for local tilt. Conventions as in Figure 4.

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TABLE 3. Site Mean Characteristic Magnetization Data from the Sites of the Builth-Llandrindod Inler in and Near the Llanelwedd Quarry

						Tilt Corrected					
				In Situ		Local		Lower Unconformity		Upper Unconformity	
Site	N	k	α_{95}	Dec	Inc	Dec	Inc	Dec	Inc	Dec	Inc
LI 7	15	759.6	1.4	144.0	43.3	203.8	50.9	176.9	42.2	174.3	71.8
LI 9	14	81.2	4.4	140.4	38.6	170.6	42.7	169.9	41.0	160.4	69.0
LI 11	15	79.2	4.3	151.9	50.7	201.6	52.3	189.3	43.3	204.7	73.1
LI 12	14	583.9	1.6	131.1	43.3	169.9	46.5	167.9	49.4	143.3	75.8
LI 13	15	59.9	5.0	140.1	40.8	172.1	45.6	171.8	42.8	162.5	71.1
LI 15	14	257.1	2.5	132.9	35.3	159.7	39.5	161.0	42.8	142.2	67.7
LI 16	8	20.0	12.7	132.8	40.5	163.4	48.4	166.1	46.6	145.4	72.7
LI 18	10	24.2	10.0	150.2	35.9	173.4	36.0	174.7	33.7	174.4	63.0
LI 19	15	77.1	4.4	143.7	36.1	170.3	38.0	170.1	37.4	163.9	65.6
BL 5	16	196.3	2.6	141.4	57.6	202.3	48.9	192.5	52.2	225.8	81.1
BL 10	14	184.9	2.9	147.2	33.3	165.1	34.6	166.7	45.1	167.1	61.9
BL 11	19	143.9	2.8	158.9	28.5	178.3	39.3	175.0	35.9	179.3	52.9
Means	12	63.5	5.5	143.0	40.6						
	12	42.4	6.7			176.4	44.5				
	12	92.5	4.5					173.3	43.0		
	12	63.5	5.5							168.1	69.9
Poles						11.8° S; 0.3° E δp=5.3 δm=8.5		12.8° S; 3.2° E δp=3.5 δm=5.6		16.2°N; 4.3° E δp=8.1 δm=9.4	

See text for explanation of the two "unconformity" tilt corrections.

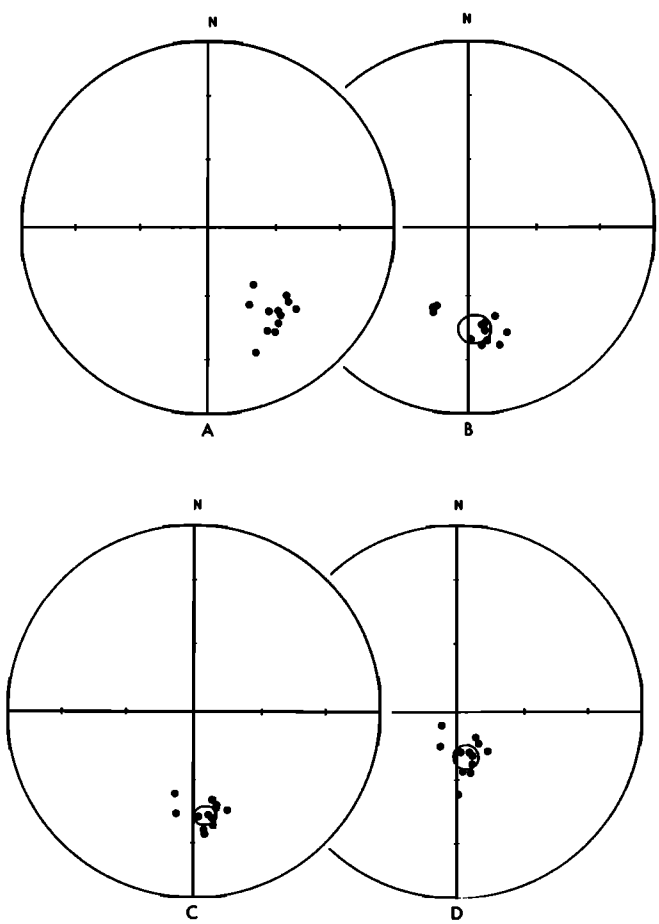


Fig. 9. Site mean characteristic directions for the southern sites in equal area projection, (*a*) before bedding correction, (*b*) after correction for local tilt, (*c*) after the lower unconformity tilt correction, and (*d*) after the upper unconformity tilt correction (see text for explanation of the "unconformity" corrections). The 95% circle of confidence about the mean is shown in some of the plots (the mean directions are omitted for clarity).

are shown after correction for local tilt in Figure 5 (left). The rock magnetic results reported by *Trench et al.* [1991] suggest that magnetite is the magnetization carrier in all of these lithologies.

Two of the southern sites were in medium-grained Newmead Sandstone, and these did not demagnetize stably. Of the remaining 16 southern sites, all in igneous units, 12 yielded a well-resolved characteristic magnetization and 10 a Late Paleozoic overprint magnetization. Site mean data on the characteristic magnetization from the 12 southern sites that yielded good results are shown in Table 3 and Figure 9. Since all the southern sites dip towards the west or west-southwest, performing a tilt correction has little affect on the overall precision parameter. *Trench et al.* [1991] found that a small-scale fold in a keratophyre flow failed the fold test. They noted that the fold may have been syn-volcanic or flow folding rather than tectonic in origin. This seems likely in light of the very convincing conglomerate test results, and the fact that the flow foliation in silicic volcanic rocks commonly exhibits primary folding. *Trench et al.* [1991] noted that the Newmead Sandstone above the erosional surface capping the volcanic pile has the same westward dips as the volcanic strata. In order to circum-

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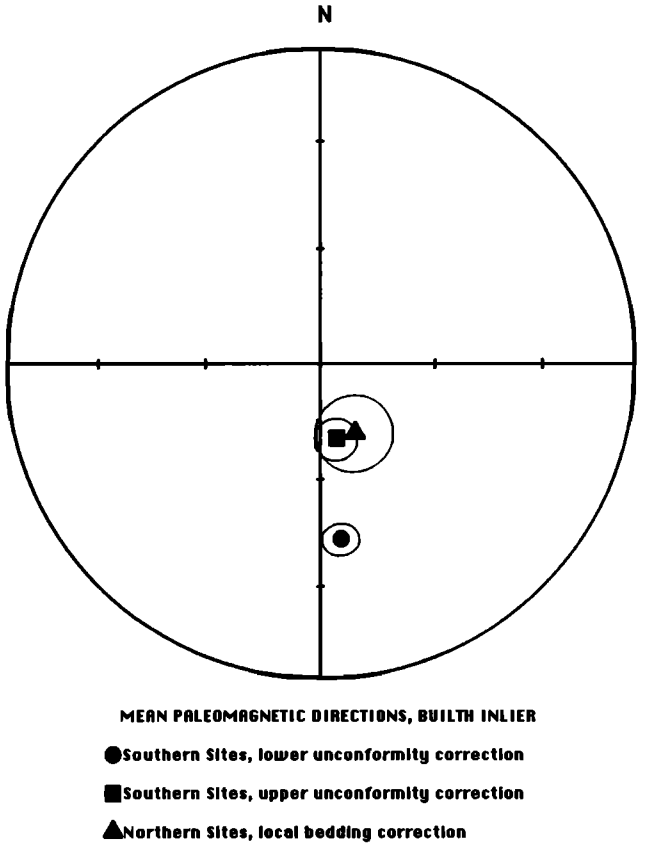


Fig. 10. Mean of site mean characteristic magnetization directions in equal area projection for the northern sites (tilt corrected), and the southern sites using the two different unconformity corrections. Note that the northern and southern mean directions agree only if the upper unconformity correction is used to correct the southern data.

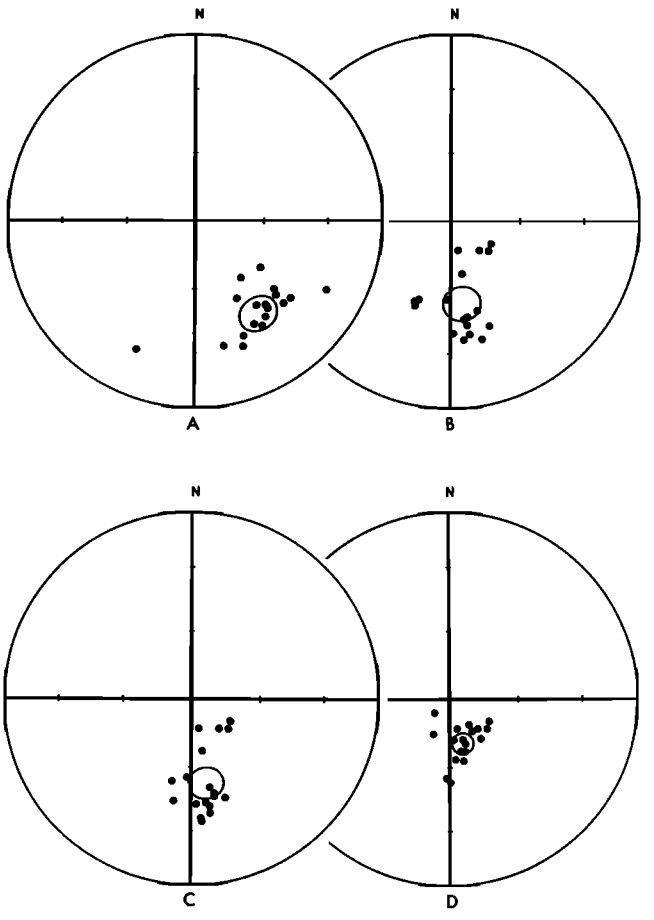


Fig. 11. Site mean characteristic directions for all Builth inlier sites in equal area projection, (*a*) before bedding correction, (*b*) after correction for local tilt, (*c*) after the lower unconformity tilt correction is applied to the southern site data, and (*d*) after the upper unconformity tilt correction is applied to the southern site data. The 95% circles of confidence about the means are shown (the mean directions are omitted for clarity). Directions cluster best when the upper unconformity correction is used (*d*).

vent the problem of flow folding in the volcanics, the attitude of nearby exposures of the Newmead was used as an estimate of paleohorizontal for the volcanic units [*Trench et al.*, 1991]. We have also used this "lower unconformity" correction to correct our site data (Table 3 and Figure 9*c*).

As noted above, the Newmead sandstone is overlain by Llandeilo marine shales with weak angular discordance, but this unconformity is not present in the northern area of the inlier [*Elles*, 1939; *Jones and Pugh*, 1949]. In addition, there is no known tectonic deformation of this age in the region. This suggests the possibility that this "upper" unconformity might be the result of early volcanotectonic deformation, or possibly primary bedding dips associated with the Llanelwedd volcanic edifice. We have therefore sought to obtain the attitude of the Llandeilo shales as an "upper unconformity" correction, under

TABLE 4. Formation Mean Directions and Paleomagnetic Poles for the Characteristic Magnetization Direction Obtained from the Builth-Llandrindod Inlier Ordovician rocks

Correction	*N*	*k*	α_{95}	Dec	Inc	Paleolat	Pole	δp	δm
In situ	18	20.3	7.9	145.5	39.6	22.5	9.7°S; 29.0°E	5.7	9.4
Local t.c.	18	20.6	7.8	172.1	53.2	33.8	3.9°S; 3.6°E	7.5	10.8
Lower unconf.*	18	23.8	7.2	169.6	52.1	32.7	4.8°S; 5.7°E	6.8	9.9
Upper unconf.**	18	57.3	4.6	163.4	70.0	53.9	16.8°N; 7.1°E	6.8	7.9

* The lower unconformity correction was applied to the southern sites only (i.e. those within and near the thick volcanic pile in the vicinity of Llanelwedd). The correction is the attitude of nearby exposures of the Newmead Sandstone, which immediately overlies the volcanic sequence.

** The upper unconformity correction was again only applied to the southern sites, and is the average bedding attitude of Llandeilo marine shales overlying the Llanelwedd volcanic pile and Newmead Sandstone (Strike=215°; Dip=33°NW).

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TABLE 5. Site Mean Secondary Magnetization Data from the Builth-Llandrindod Inlier

Site	N	k	α_{95}	in situ Dec	in situ Inc	tilt corrected Dec	tilt corrected Inc
LI 6	8	75.2	6.4	197.3	-31.1	221.5	-9.8
LI 9	14	114.4	3.7	179.1	-17.5	166.0	-24.0
LI 10	14	100.1	4.0	178.6	-22.1	158.8	-29.2
LI 11	12	110.9	4.1	179.3	-21.4	166.0	-17.1
LI 13	14	33.7	6.9	172.1	-2.7	169.3	-6.7
LI 16	12	87.6	4.7	181.5	-20.2	167.9	-27.2
LI 18	10	29.6	9.0	181.8	-11.2	173.3	-19.7
LI 19	12	119.5	4.0	177.6	-9.6	169.3	-17.8
BL 7	13	23.6	8.7	196.1	-24.8	190.8	-35.5
BL 9	14	563.7	1.7	178.2	-22.6	169.3	0.5
BL 13	14	57.4	5.3	185.3	-23.0	175.6	-19.3
Means	11	56.6	6.1	182.2	-18.9		
	11	17.9	11.1			174.9	-19.5
Pole (in situ correction)				47.7° S; 353.8° E $\delta p = 3.3°$ $\delta m = 6.4°$			

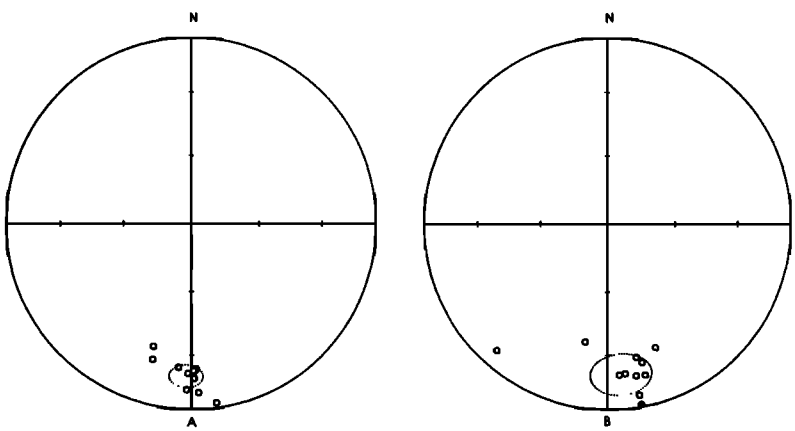


Fig. 12. Site mean secondary magnetization directions in equal area projection (a) before, and (b) after correction for local tilt, with 95% circles of confidence about the means shown (the mean directions are omitted for clarity).

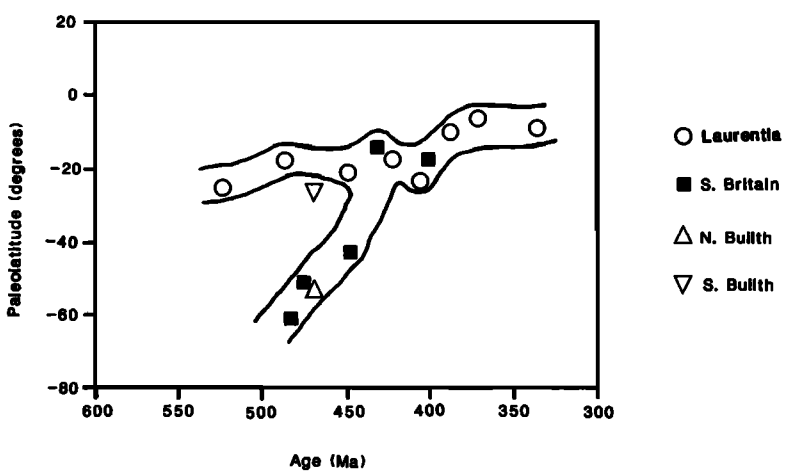


Fig. 13. Paleolatitudes obtained in this study from southern and northern Builth as indicated, along with other results from southern Britain (listed in Table 6). These paleolatitudes are compared with reference data from Laurentia for a reference point in southern Britain with the Atlantic Ocean closed. [after *Van der Voo*, 1992].

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TABLE 6. Key to Figure 13

Rock Unit [Reference]		Pole Latitute, deg	Pole Longitude, deg	dp/dm, deg
Treffgarne Volcanics [*Trench et al.*, 1992]	L. Ord.	56N	306E	9/10
Stapeley Volcanics [*McCabe and Channell*, 1990]	M. Ord.	26.6N	36.1E	6.9/8.2
N. Builth (seds/volcs/intrusives) [this paper]	M. Ord.	17.9N	12.9E	15/17
S. Builth (volcs/intrusives) [this paper]*	M. Ord.	12.8 S	3.2E	3.5/5.6
Borrowdale Volcanics [*Channell and McCabe*, 1992]	U. Ord.	8.1N	6.2E	8.2/10.2
Mendip inlier volcanics [*Torsvik and Trench*, 1992]	L. Sil.	13N	271E	5/9
Lower Old Red Sandstone [*Channell et al.*, 1992b]	L. Dev	7.3S	306.7E	5.4/9.5

Selected Early and Middle Paeozoic paleomagnetic results from England and Wales.
*Lower unconformity correction.

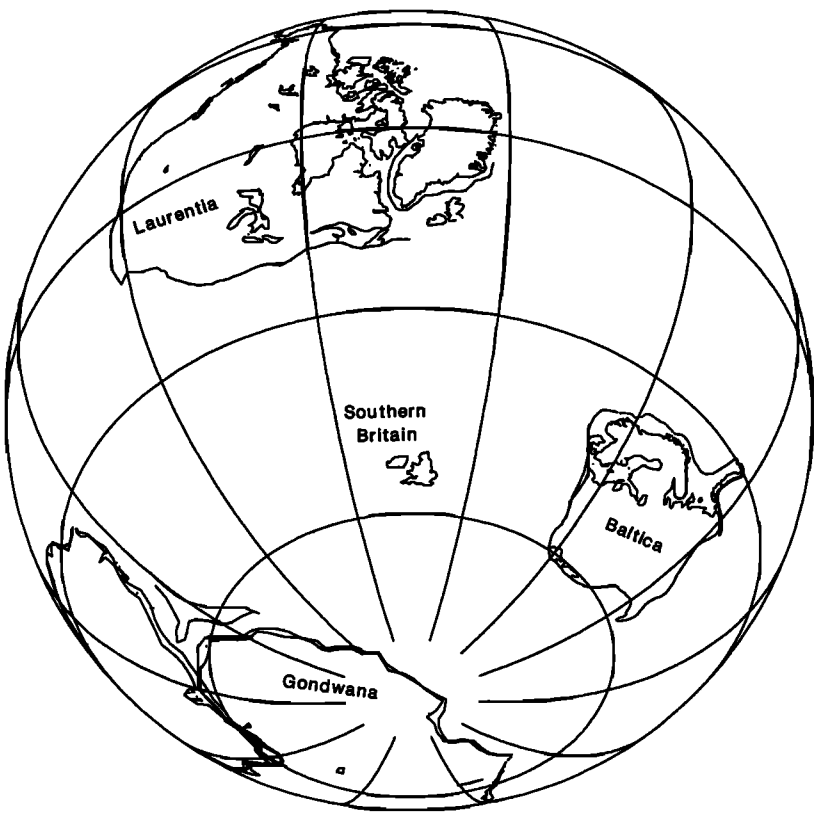


Fig. 14. Paleogeographic map showing the Middle Ordovician latitudinal relationships between some major tectonic elements and southern Britain according to data from the six northern Builth inlier sites (this study). Laurentia was positioned according to the mean Middle Ordovician pole for North America from *Van der Voo* [1990], and Gondwana according to the Middle-Late Ordovician pole for Gondwana in *Van der Voo* [1988]. Baltica was positioned with the Lower to Middle Ordovician pole from the Orthoceras Limestons obtained by *Torsvik and Trench* [1991].

the working hypothesis that this represents the true paleohorizontal associated with the volcanic units. Unfortunately, this contact is not well exposed. *Elles* [1939, p. 408-409] describes a quarry exposure of this unconformity near the Cwm Amliw farm, where the Llandeilo shales have an attitude of Strike = 200°, Dip = 30°W. *Jones and Pugh* [1949, p. 96] note that the outcrop pattern of the contact in the area between the Carneddau and Wern-to faults demonstrates that the contact is with angular discordance. The *Institute of Geological Sciences* [1977] map of the inlier shows that the attitude of the Llandeilo shales is quite consistent to the west of the Llanelwedd volcanic pile exposures, except where it is disturbed near dolerites. The average attitude of undisturbed Llandeilo strata from data given on the map is Strike = 215, Dip = 33°NW. This paleohorizon has been used to correct the southern site data as an "upper unconformity" correction (Table 3; Figure 9*d*).

Comparison of the Northern and Southern Site Data

Figure 10 shows a comparison of the mean characteristic magnetization direction from the northern sites with the mean of both the "lower unconformity" and "upper unconformity" corrected data from the southern sites. The "upper unconformity" corrected southern mean direction is not distinguishable from the tilt corrected northern mean. However, the "lower unconformity" corrected southern mean is significantly shallower

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(Figure 10). Figure 11 shows the mean characteristic directions from all sites. Figure 11*c* shows the tilt corrected northern site means and the southern site means corrected according to the lower unconformity correction. Figure 11*d* is similar to Figure 11*c*, except that the southern site data are corrected according to the upper unconformity correction. The means of all 18 site means, associated statistics, and paleomagnetic poles for the different tilt correction models are shown in Table 4. Application of the upper unconformity correction to the southern site data results in the best clustering of directions, significantly better than either the local tilt correction or the lower unconformity correction at the 99% confidence level.

THE SECONDARY MAGNETIZATION

Site mean paleomagnetic data for the secondary magnetization is listed in Table 5 and depicted graphically in Figure 12. The directions cluster significantly better (99% confidence level) before the local tilt correction is applied, indicating a post-Ordovician magnetization age. The mean direction is close to the Late Carboniferous expected direction for the site, and the exclusively reversed polarities suggest magnetization during the Kiaman reversed superchron. We previously noted [*McCabe and Channell*, 1991] that this Late Paleozoic magnetization overprint is found mainly in samples from the Llanelwedd quarry, which lies within a fault zone [*Woodcock*, 1987]. This suggests that the faults might have served to focus the flow of remagnetizing fluids. By analogy with studies on Late Paleozoic fluid-mediated remagnetization in North America [reviewed by *McCabe and Elmore*, 1989], we suggest that the partial remagnetization of some Builth inlier rocks occurred as a consequence of the mobilization of metamorphic or basinal fluids in the foreland to the active Hercynian orogen.

DISCUSSION AND CONCLUSIONS

Our data indicate that results from the Llanelwedd quarry (the Builth southern sites) are not only anomalous with respect to those from the Shelve inlier [*McCabe and Channell*, 1990], but that they are also anomalous with respect to the data from the central and northern areas of the Builth inlier itself. We also find them to be anomalous in the context of all the best-documented Early and Middle Paleozoic results from southern Britain [see *Channell et al.*, 1992a]. Figure 13 shows paleolatitudes obtained in this study along with other pertinent results from England and Wales (data listed in Table 6) compared to the paleolatitudes expected for Scotland based on the Laurentian database. Data from northern Builth fall within the path defined by the other data points indicating progressive, rapid closure of the Iapetus Ocean from about Early Ordovician to mid-Silurian. However, data from southern Builth appear to be anomalous with respect to the other data. We therefore regard the southern Builth data with suspicion, and consider the northern Builth and Shelve data to be most representative of the Middle Ordovician field for southern Britain. Figure 14 shows the paleogeographic implications of the northern Builth data for the Iapetus-bordering continents during the Middle Ordovician.

The origin of the anomalously shallow inclinations from southern Builth is difficult to explain. Application of the upper unconformity correction to the paleomagnetic data from the southern sites provides one paleomagnetic solution, since it reconciles the discrepant data from the northern and southern sites, and results in good agreement with the approximately coeval result from the Shelve inlier volcanics [*McCabe and Channell*, 1990]. However, use of this correction implies that the volcanic strata were not horizontal at the time the magnetization was acquired, and this may not be geologically reasonable. Figure 15 shows average bedding poles from the stratified units in the vicinity of Llanelwedd quarry, in coordinates corrected according to the upper unconformity correction. The great circle plotted in the figure shows the inferred attitude of the strata at the time the volcanics acquired their magnetization, provided that the upper unconformity is a good estimate of the paleohorizontal. This implies a southerly dip of some 25° at the time of magnetization.

A problem with the use of this paleo-horizon lies in the conglomerate test results showing that the volcanic magnetization was acquired prior to deposition of the Newmead Sandstone, which has about the same present attitude as the volcanics. This inevitably implies that the Newmead Sandstone was deposited with primary dips of some 25°. This may be possible since this dip angle is less than the angle of repose for medium-grained sand, and primary basinward dips have been previously reported for sandstones deposited adjacent to wave cut cliffs in the Paleozoic "Baraboo Islands" of Wisconsin [*Dalziel and Dott*, 1970]. However, we consider this unlikely, given that the Newmead Sandstone exhibits rare ripple cross lamination, which is inconsistent with primary dips of more than a few degrees. In a previous discussion of these data [*Channell et al.*, 1992a] we favored repeated episodes of

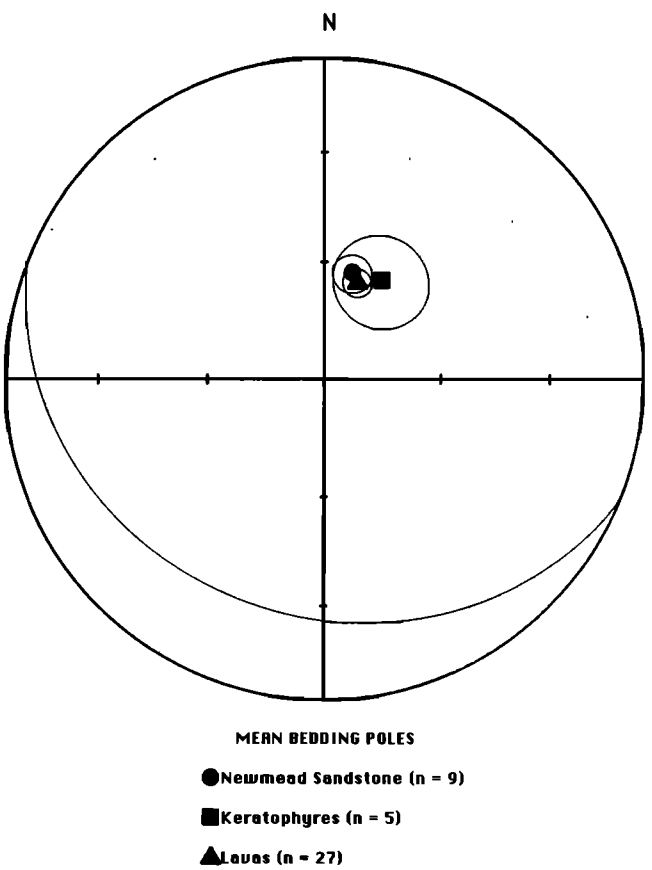


Fig. 15. Mean bedding poles from volcanic and sedimentary rock units in the southern area of the inlier, shown here corrected according to the attitude of the overlying Llandeilo shales (i.e. the upper unconformity correction). The plotted great circle is the inferred attitude of the units at the time of magnetization, provided that the use of the upper unconformity correction is appropriate. Equal area projection.

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volcano tectonic tilting prior to deposition of the Llandeilo shales. If this were the case, the angular relationship between lavas and the overlying Llandeilo shales at the time of magnetization would be unknown, and the resolution of the northern/southern discrepancy by the upper unconformity correction would be fortuitous. Although it is possible that future sedimentologic and/or volcanological studies may justify the use of the upper unconformity correction to the quarry data, we consider that the geological difficulties it presents are a compelling reason to discard this correction for now, despite of it's success in reconciling the discrepant paleomagnetic data.

Another possible origin for the anomalous nature of the southern Builth data is that the Llanelwedd volcanic pile was deposited too rapidly for secular variation to be averaged out. Although this idea is impossible to test, a large database for Cenozoic mafic volcanics from the western United States suggests that this solution is a reasonable one. For example, *Calderone et al.* [1990] review data from 179 Miocene lava flows, but show that only 65 time-distinct VGPs are present in the data. This suggests that volcanism tends to be episodic, with multiple lava flows recording essentially the same VGP. In another study, *Geissman et al.* [1990] show that a highly anomalous field (an excursion) of Brunhes age is present, virtually unchanged, in a sequence of at least six separate lava flows in New Mexico. These Cenozoic results indicates that it is not unreasonable to suggest that the approximately 200 m of spilites and keratophyres exposed in the Llanelwedd quarry may have been extruded more rapidly than the ~10 ka necessary to average out secular variation.

For the present time, we suggest that the mean of the northern six sites (Table 2; D=153.8°, I=69.8°; k=45.8; α_{95}=10°) probably provides the best estimate of the Middle Ordovician paleomagnetic field of the Builth inlier, since the bedding correction of at least some of these sites (i.e. the sediments and tuffs) is almost certainly well-determined, and because of the excellent agreement of the paleolatitude with the one obtained from the Shelve inlier (54° versus 51°). Although this is not an entirely satisfactory solution, we think that although the characteristic magnetization in the southern part of the inlier was acquired early, the southern Builth data are not representative of the geocentric axial dipole field for southern Britain and that they should not be used to estimate the Middle Ordovician paleolatitude for this terrane.

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